Hsuan-Fu (Jeff) Hsu

AI / ML Engineering | Python, LLM Agents, Model Evaluation and Cloud Deployment

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science | GPA: 3.6/4.0 — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science | GPA: 3.5/4.0 — Sept 2020 - June 2024

SKILLS

Languages: Python, SQL, C++, C#, TypeScript, JavaScript

ML & Data: NumPy, PyTorch, TensorFlow, Sentence Transformers, tsfresh, signal preprocessing, feature selection, LOSO validation

Agents & APIs: LangChain, LangGraph, MCP, FastAPI, ASP.NET Core, Node.js, Express

Cloud & Databases: Docker, Google Cloud Run, SQL Server, SQLite

Engineering: pytest, xUnit, Vitest, integration testing, GitHub Actions CI, Git, Linux

EXPERIENCE

National Taipei University (NTPU) — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- **Co-designed the experimental protocol and signal pipeline** for a 3-person research team, processing noninvasive PPG signals from 30 participants across 2 near-infrared wavelengths with Bessel filtering and tsfresh feature extraction.
- **Led model selection** across 5 ML model families and 4 DL architectures using **leave-one-subject-out (LOSO) cross-validation**; the team's linear regression model achieved the best result of **10.3% MAPE** with 10 selected features.
- Earned individual NSTC research funding and 4th place in NTPU's project competition with a team of 3.

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Independently designed and built a government birth grant system with **ASP.NET Core MVC**, now deployed across 29 household registration offices, with application status tracking for applicants and staff.
- **Implemented incremental data synchronization** across all 29 offices, automatically transferring only changed data instead of requiring staff to manually upload complete datasets.
- Optimized **SQL Server** data processing and storage, reducing database latency by **at least 50%** and overall resource usage by **at least 30%** compared with the previous implementation.

PROJECTS

Scientific Retrieval Fine-tuning [GitHub] — Oct 2026
Python, PyTorch, Sentence Transformers, MPS

- Designed a 3-arm controlled comparison (original, random-negative, hard-negative) fine-tuning **MiniLM** with **TripletLoss** on Apple Silicon MPS, matched across loss, batch size, learning rate, and 3 fixed training seeds.
- Built a leakage-checked 649/160 train/validation split (490 dedup groups) and a false-negative review process over mined hard negatives before training; verified with **53 unit tests**.
- Hard-negative fine-tuning improved held-out BEIR SciFact test **nDCG@10 by 3.2%** and **Recall@10 by 2.1%** over matched random-negative fine-tuning, consistent across all 3 seeds.

LLM Agent with MCP Tools — Aug 2026 - Present
Python, LangGraph, FastAPI

- Built a Python LLM agent with **LangGraph** and NVIDIA NIM, integrating 3 MCP servers for web search, file operations, and custom calculation tools.
- Added session memory and per-session locks within one service instance, plus tool retries, input validation, and sanitized errors for failed calls.
- Deployed a **FastAPI** service with Docker on Google Cloud Run; added pytest tests for configuration, tool validation, and error handling.

Promptfoo Agent Retry Validation [GitHub] — Oct 2026
TypeScript, Vitest

- Added nested-path error localization to the OpenAI Agents retry-policy validator in a personal promptfoo fork, pinpointing the exact invalid field instead of a generic error message.
- Wrote **81 new test cases (195 total passing)** covering config parsing, real-SDK retry behavior, and provider/inline-agent edge cases.
- Fixed **6 error-localization regressions** in a reproducible before/after comparison, with no change to existing validation or retry behavior.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, accepted for publication, Aug 2026.